Windows PowerShell

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Windows PowerShell
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PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D5_LangChain_Project> python W6D5_langchain_project.py

======================================================================
TASK 1: LANGCHAIN CHAIN
======================================================================

Input 1: What is LangChain?
Output 1: LangChain is an open-source framework for building data pipelines and workflows. It provides a set of tools and libraries t
hat enable developers to create modular, scalable, and maintainable data processing applications. LangChain allows users to define, e
xecute, and manage complex data workflows using a combination of programming languages and domain-specific languages (DSLs).

Input 2: What is an LLM?
Output 2: An LLM (Large Language Model) is a type of artificial intelligence model that uses natural language processing (NLP) to und
erstand and generate human-like language. It's trained on vast amounts of text data to learn patterns, relationships, and context, al
lowing it to produce coherent and relevant responses. LLMs are designed to assist with tasks such as language translation, sentiment
analysis, and text summarization.

Input 3: What is a prompt template?
Output 3: A prompt template is a pre-defined format or structure for creating text prompts that can be used to interact with large la
nguage models (LLMs). It typically includes placeholders for specific information, such as entities or attributes, which are then fil
led in by the user when generating a specific request. Prompt templates help ensure consistency and clarity in input, making it easie
r for LLMs to understand the intended meaning.

Input 4: What is a vector database?
Output 4: A vector database is a type of database that stores and indexes data as vectors in a multidimensional space. This allows fo
r efficient querying and similarity search of the stored data, often used in applications such as recommendation systems, image retri
eval, and text search. Vector databases are typically optimized for fast lookup, nearest neighbor search, and other distance-based qu
eries.

Input 5: What is RAG?
Output 5: RAG stands for Relational Algorithmic Graph. It's a type of graph-based architecture used in natural language processing (N
LP) to efficiently store and query large amounts of structured data, such as knowledge graphs or databases. In the context of NLP, RA
Gs are designed to enable fast and scalable querying of structured information from large datasets, making them suitable for applicat
ions like question answering and text summarization.
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TASK 2: CONVERSATION BUFFER MEMORY
======================================================================
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D5_LangChain_Project\W6D5_langchain_project.py:94: LangChainDeprecationWar
ning: The class `ConversationBufferMemory` was deprecated in LangChain 0.3.1 and will be removed in 2.0.0. Use `langchain.agents.crea
te_agent` instead. For agents that need to remember prior interactions, use `create_agent` with checkpointing or the `Store` API. See
 https://docs.langchain.com/oss/python/langchain/short-term-memory and https://docs.langchain.com/oss/python/langchain/long-term-memo
ry
  memory = ConversationBufferMemory(

Turn 1
User: My name is Sanjana.
Assistant: Nice to meet you, Sanjana! I don't have any information about you yet. How are you doing today? Is there something on your
 mind that you'd like to talk about or ask about?

Turn 2
User: I am learning LangChain during my internship.
Assistant: That's great that you're learning LangChain during your internship, Sanjana! LangChain is an open-source framework for bui
lding conversational AI models. What specifically are you focusing on while learning LangChain? Are you working on a project or tryin
g to improve your understanding of the library and its applications?

Turn 3
User: What topic am I learning?
Assistant: You're currently learning about the LangChain library, which is an open-source framework for building conversational AI mo
dels. Is that correct?

Turn 4
User: What is my name?
Assistant: Your name is Sanjana.

Turn 5
User: Can you summarize what you know about this conversation?
Assistant: Here's a summary of our conversation:

* You introduced yourself as Sanjana and mentioned that you're learning LangChain during your internship.
* I asked if there was something on your mind that you'd like to talk about or ask about, but you didn't provide any specific topic y
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* We discussed what you're focusing on while learning LangChain (although you said "What topic am I learning?" initially).
* You corrected me by asking what topic you were learning again, and then asked if your name was correct.
* Finally, you asked me to summarize our conversation.

Let's continue the conversation from here! Is there anything specific you'd like to discuss or ask about LangChain?

FINAL STORED CONVERSATION HISTORY:
Human: My name is Sanjana.
AI: Nice to meet you, Sanjana! I don't have any information about you yet. How are you doing today? Is there something on your mind t
hat you'd like to talk about or ask about?
Human: I am learning LangChain during my internship.
AI: That's great that you're learning LangChain during your internship, Sanjana! LangChain is an open-source framework for building c
onversational AI models. What specifically are you focusing on while learning LangChain? Are you working on a project or trying to im
prove your understanding of the library and its applications?
Human: What topic am I learning?
AI: You're currently learning about the LangChain library, which is an open-source framework for building conversational AI models. I
s that correct?
Human: What is my name?
AI: Your name is Sanjana.
Human: Can you summarize what you know about this conversation?
AI: Here's a summary of our conversation:

* You introduced yourself as Sanjana and mentioned that you're learning LangChain during your internship.
* I asked if there was something on your mind that you'd like to talk about or ask about, but you didn't provide any specific topic y
et.
* We discussed what you're focusing on while learning LangChain (although you said "What topic am I learning?" initially).
* You corrected me by asking what topic you were learning again, and then asked if your name was correct.
* Finally, you asked me to summarize our conversation.

Let's continue the conversation from here! Is there anything specific you'd like to discuss or ask about LangChain?

======================================================================
TASK 3: LANGCHAIN AGENT WITH 2 TOOLS
======================================================================

Agent Task 1: Use the web search tool to tell me what LangChain is.
Agent Output 1: Is there a specific aspect of LangChain you'd like to know more about?
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* You introduced yourself as Sanjana and mentioned that you're learning LangChain during your internship.
* I asked if there was something on your mind that you'd like to talk about or ask about, but you didn't provide any specific topic y
et.
* We discussed what you're focusing on while learning LangChain (although you said "What topic am I learning?" initially).
* You corrected me by asking what topic you were learning again, and then asked if your name was correct.
* Finally, you asked me to summarize our conversation.

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======================================================================
TASK 3: LANGCHAIN AGENT WITH 2 TOOLS
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Agent Task 1: Use the web search tool to tell me what LangChain is.
Agent Output 1: Is there a specific aspect of LangChain you'd like to know more about?
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Agent Task 2: Calculate 125 * 8 + 50.
Agent Output 2: The result of the calculation is 1050.

Agent Task 3: Use the web search tool to find the stub information about Ollama, then calculate 45 / 5 and include both results in yo
ur answer.
Agent Output 3: The Ollama stub information indicates that it's a local AI model runtime for running large language models on persona
l computers.

The calculation 45 / 5 results in 9.0.

======================================================================
W6D5 COMPLETION SUMMARY
======================================================================
Task 1: Chain created and tested with 5 inputs.
Task 2: ConversationBufferMemory tested across 5 turns.
Task 3: Agent created with 2 tools and tested with 3 tasks.
Required stack: LangChain + Ollama.
Output evidence is printed in the terminal.
W6D5 practical implementation completed.
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D5_LangChain_Project>
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Show hidden icons

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